

PHYS 142/242 Assignment 1: Harmonic Oscillator Path Integral

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$$x_{\text{start}} = 0.89$$

Units: $m = \omega = \hbar = 1$, $T_0 = 2\pi$. Grid: $N_D + 1 = 601$ points on $[-4, 4]$, $\Delta x = 8/600$; time step $\epsilon = T_0/128$. All code is in `assignment1.ipynb`. It writes every figure below, `animation.gif`, and `results.json`. States are evolved as $\psi_{n+1} = \Delta x K_\epsilon \psi_n$, where $(K_\epsilon)_{ba} = K(x_b, \epsilon; x_a, 0)$ is Eq. (4). The state is never renormalized.

Problem A: $K_{8\epsilon}$

Using Eq. (5) with $N = 8$, $K_{8\epsilon} = (\Delta x)^7 K_\epsilon^8$ (one factor of Δx for each of the 7 intermediate integrals). Truncated NumPy output (`print(K_8eps)`):

```
[[-0.05955846+0.0178667j -0.0757611 +0.0217024j -0.07967385+0.05414053j
... -0.46505982-0.30932634j -0.35439299-0.42735575j
-0.21687407-0.50917202j]
[-0.0757611 +0.0217024j -0.0896081 +0.03330227j -0.08308486+0.06724787j
... -0.53404108-0.16837676j -0.46141968-0.31131639j
-0.35439299-0.42735575j]
[-0.07967385+0.05414053j -0.08308486+0.06724787j -0.06226964+0.09303051j
... -0.56407376-0.01040964j -0.53404108-0.16837676j
-0.46505982-0.30932634j]
...
[-0.46505982-0.30932634j -0.53404108-0.16837676j -0.56407376-0.01040964j
... -0.06226964+0.09303051j -0.08308486+0.06724787j
-0.07967385+0.05414053j]
[-0.35439299-0.42735575j -0.46141968-0.31131639j -0.53404108-0.16837676j
... -0.08308486+0.06724787j -0.0896081 +0.03330227j
-0.0757611 +0.0217024j ]
[-0.21687407-0.50917202j -0.35439299-0.42735575j -0.46505982-0.30932634j
... -0.07967385+0.05414053j -0.0757611 +0.0217024j
-0.05955846+0.0178667j ]]
```

As expected, the matrix is symmetric: $K(x_b, x_a) = K(x_a, x_b)$. Check: acting on ψ_0 , $\Delta x K_{8\epsilon} \psi_0$ agrees with $\Delta x K(8\epsilon) \psi_0$ from Eq. (4) at $t = 8\epsilon$ to a relative error of 1.62×10^{-4} . Comparing the two matrices element by element is not a meaningful test. A point-source kernel spreads over every x and every momentum, so its entries depend on where the box ends and on the grid's momentum cutoff $\pi/\Delta x$. On a smooth, localized state those effects drop out.

Problem B: $\langle x \rangle$ over one period

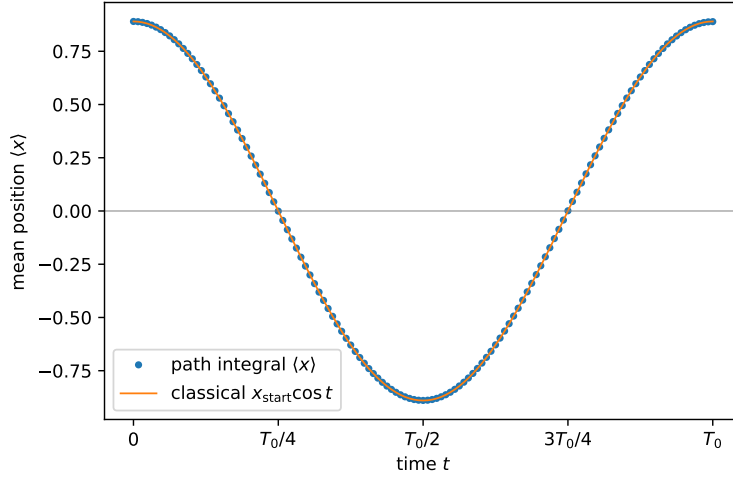


Figure 1: $\langle x \rangle = \Delta x \sum_i x_i |\Psi_t(x_i)|^2$ over one period (dots), with the classical trajectory $x_{\text{start}} \cos t$ (line).

As Ehrenfest's theorem requires for a quadratic potential, $\langle x \rangle$ follows the classical trajectory exactly. The maximum deviation is 1.04×10^{-3} , and after one period $\langle x \rangle(T_0) = 0.8892$. The small residual comes from the hard walls at $x = \pm 4$. Around $t = T_0/4$ and $3T_0/4$ the packet's tails reach the walls and are cut off. The norm drops to 0.99940 by $t = T_0$. Repeating the run with walls at ± 6 (same Δx) brings the norm loss down to $\sim 3 \times 10^{-8}$.

Problem C: $\langle E \rangle$, $\langle K \rangle$, $\langle V \rangle$

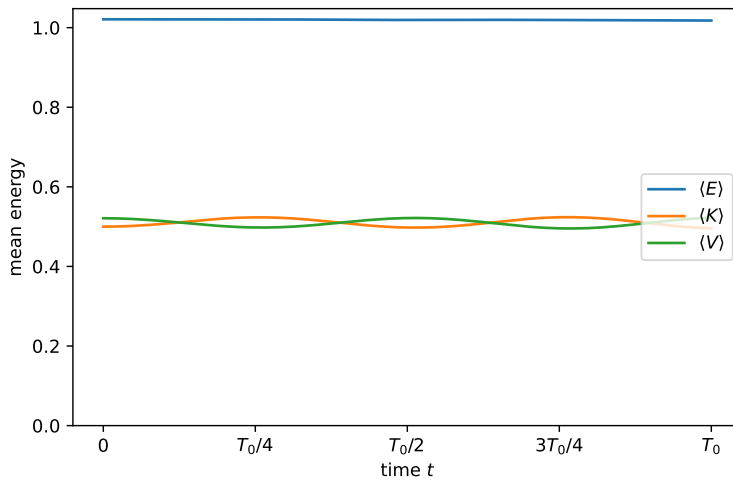


Figure 2: Mean total, kinetic, and potential energy over one period. $\langle V \rangle = \Delta x \sum_i \frac{1}{2} x_i^2 P_t(x_i)$ and $\langle K \rangle = \frac{1}{2} \Delta x \sum_i |\partial_x \Psi_t(x_i)|^2$ (Eq. 8), with the derivative taken by central differences.

Analytically, the initial Gaussian has $\langle K \rangle(0) = \alpha/4 = 0.50000$ and $\langle V \rangle(0) = \frac{1}{2}x_{\text{start}}^2 + \frac{1}{4\alpha} = 0.52105$, so $\langle E \rangle = 1.02105$. Numerically, $\langle K \rangle(0) = 0.49991$ and $\langle V \rangle(0) = 0.52105$, and $\langle E \rangle$ stays in $[1.01785, 1.02144]$: conserved to 0.31%. That small drift is again wall leakage: the probability lost near $|x| = 4$ carries $V \approx 8$ per unit probability. With walls at ± 6 the drift drops to $\sim 10^{-4}$.

$\langle K \rangle$ and $\langle V \rangle$ trade energy back and forth at frequency 2ω . The reason is that $\alpha = 2 \neq m\omega/\hbar = 1$: the initial state is narrower than the ground state (a squeezed state). The width therefore “breathes” twice per period. The position variance is $1/(2\alpha) = 0.25$ at $t = 0, T_0/2, T_0$ and $\alpha/2 = 1$ at $t = T_0/4, 3T_0/4$ (numerically 0.25 and 0.996). This breathing sits on top of the classical exchange $\frac{1}{2}x_{\text{start}}^2 \cos^2 t \leftrightarrow \frac{1}{2}x_{\text{start}}^2 \sin^2 t$, which also has frequency 2ω .

Problem D: amplitudes at $t = mT_0/16$

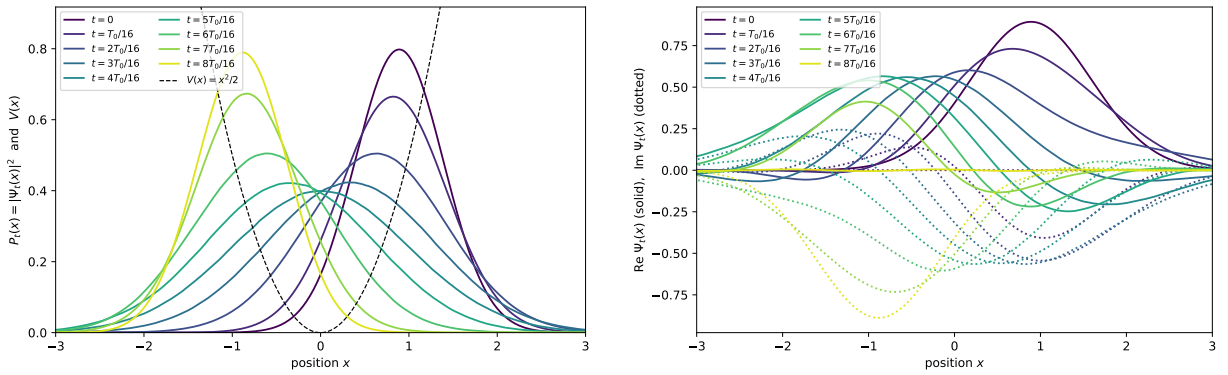


Figure 3: Left: $P_t(x) = |\Psi_t(x)|^2$ at $t = mT_0/16$, $m = 0, \dots, 8$, with $V(x) = x^2/2$ superimposed. Right: $\text{Re } \Psi_t$ (solid) and $\text{Im } \Psi_t$ (dotted) at the same times.

Over half a period the packet moves from x_{start} to $-x_{\text{start}}$, as in Problem B. It broadens to its maximum width at $t = T_0/4$ (where it is centered at $x = 0$) and then refocuses to its original width at $t = T_0/2$. In the right panel, the phase develops a linear term $e^{i\langle p \rangle x}$ while the packet moves and a quadratic chirp while it breathes. At $t = T_0/2$ the state is purely imaginary. It equals $-i \Psi_0(-x)$ (numerically to $< 1\%$, the difference being wall leakage), because $e^{-iHT_0/2} = -i \hat{\Pi}$ with $\hat{\Pi}$ the parity operator.

Problem E: animation

The animation is `animation.gif`, submitted with the code. It has 129 frames, one per time step $\epsilon = T_0/128$ from $t = 0$ to T_0 . It shows $\text{Re } \Psi$, $\text{Im } \Psi$, and $|\Psi|$ together with $V(x)$. Every 16th frame is shown below.

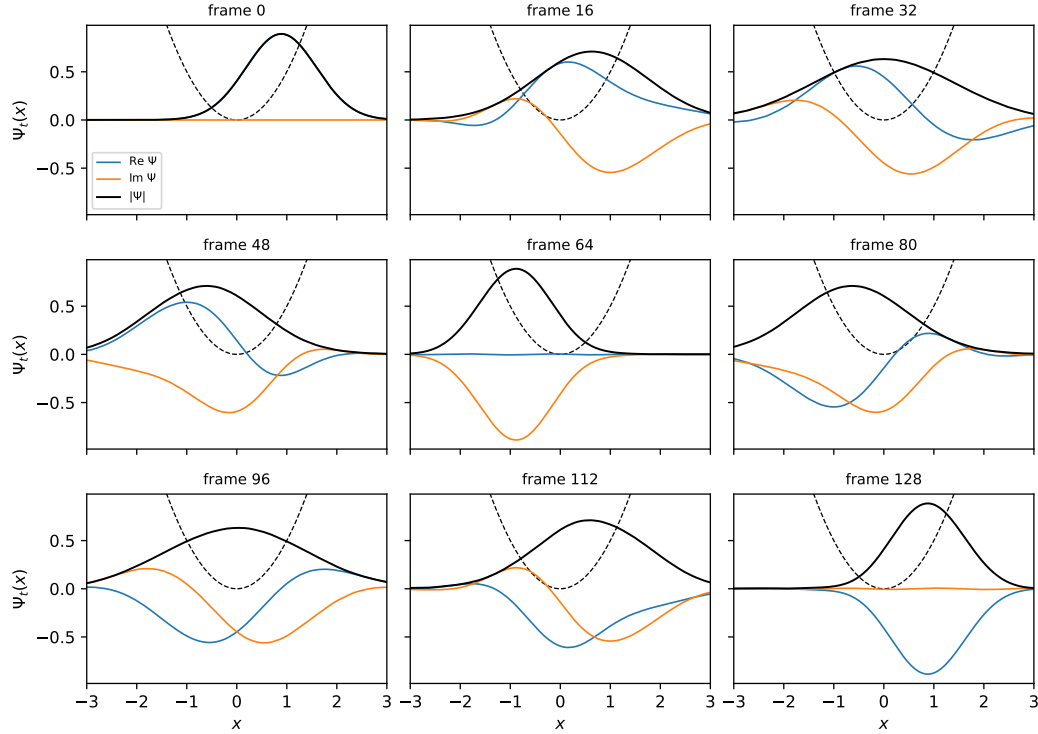


Figure 4: Frames 0, 16, ..., 128 of `animation.gif`. After one full period, $|\Psi|$ returns to its initial shape and $\Psi(T_0) = -\Psi_0$, since $e^{-iHT_0} = e^{-2\pi i(n+1/2)} = -1$ for every eigenstate.

Problem F (bonus): norm without renormalization

Prediction (made before running). The phase of K_ϵ is $\phi = [(x_a^2 + x_b^2) \cos \epsilon - 2x_a x_b] / (2 \sin \epsilon)$. As a function of x_a it oscillates with local wavenumber

$$k(x_a, x_b) = \left| \frac{\partial \phi}{\partial x_a} \right| = \frac{|x_a \cos \epsilon - x_b|}{\sin \epsilon} \lesssim \frac{8}{\sin \epsilon} \quad \text{on } [-4, 4].$$

The sum over x_a can only represent the integral if Δx resolves these oscillations. A Nyquist-type estimate (phase step $k\Delta x < \pi$) gives $8\Delta x / \sin \epsilon \approx 0.28$ for (i), 8.7 for (ii), and 8.7 for (iii). So I predicted that **(ii) and (iii) fail and (i) works**. The main run ($\epsilon = T_0/128$, $N_D = 600$) gives 2.2 and is fine. Note the counterintuitive direction: a *smaller* time step makes things worse, because K_ϵ oscillates faster as $\epsilon \rightarrow 0$, where it approaches a δ -function.

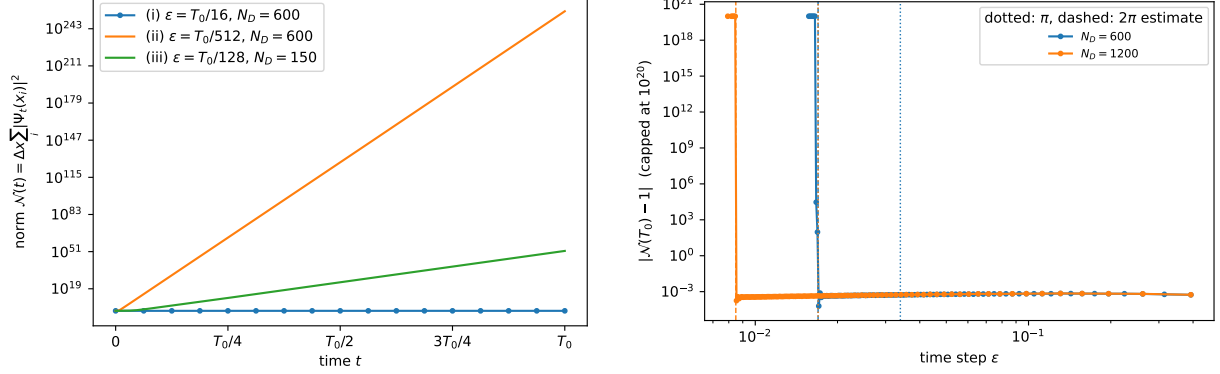


Figure 5: Left: $\mathcal{N}(t)$ for the three cases (log scale). Right: $|\mathcal{N}(T_0) - 1|$ after one period vs. $\epsilon = T_0/n$, for $N_D = 600$ and 1200 . Dotted lines: phase step π per Δx . Dashed lines: phase step 2π per Δx . Values are capped at 10^{20} for plotting.

Results. The prediction holds.

- Case (i) stays at $\mathcal{N}(T_0) = 0.99946$; the only loss is the wall leakage from Problem B.
- Case (ii) reaches $\mathcal{N}(T_0) = 1.1 \times 10^{258}$.
- Case (iii) reaches $\mathcal{N}(T_0) = 4.1 \times 10^{51}$.

The two failing runs grow geometrically: a straight line on the log plot, by a factor ≈ 3.2 and ≈ 2.5 per step. Once the kernel is undersampled, $\Delta x K_\epsilon$ is no longer a unitary matrix. Repeated application then amplifies its eigenvector with $|\lambda| > 1$ exponentially.

Explanation and smallest time step. The π criterion turns out to be conservative by a factor of 2. Poisson summation explains why: $\Delta x \sum_j f(x_j) = \sum_m \int f(x) e^{2\pi i m x / \Delta x} dx$. For $f = K_\epsilon \Psi$, the alias terms ($m \neq 0$) oscillate rapidly and cancel, unless they have a stationary point. That happens when k reaches $2\pi/\Delta x$ somewhere on the grid. The condition for the method to work is therefore

$$\frac{8 \Delta x}{\sin \epsilon} < 2\pi \implies \epsilon_{\min} \approx \frac{4\Delta x}{\pi} = \frac{32}{\pi N_D}.$$

For $N_D = 600$ this gives $\epsilon_{\min} \approx 0.017 \approx T_0/370$. The scan finds the first failure at $\epsilon = T_0/372$.

Doubling N_D halves Δx , which halves $\epsilon_{\min}$. The prediction is $T_0/740$ for $N_D = 1200$, and the scan finds $T_0/744$. In other words, a finer time step must be paired with a proportionally finer grid: $\epsilon_{\min} \propto \Delta x \propto 1/N_D$. There is no lower limit on ϵ from accuracy, since K_ϵ is exact for any ϵ . The only approximation is the discretization of x .

AI use appendix

[To be written.]